

Assignment 14

Module -2- Linux server - Operate running systems

1. View running processes with ps.

- The ps command is used to display currently running processes in Linux.

- ❖ Command

- Ps

- ❖ **Why it is Used?**

Shows:

- Running processes
 - Process ID (PID)
 - Terminal information

- ❖ **Why ps is Important?**

It helps administrators:

- Monitor running services
 - Find process IDs
 - Troubleshoot system issues

2. Terminate processes with kill.

- The kill command is used to stop or terminate processes.

- ❖ **Why Use -9?**

- Sends SIGKILL signal
 - Immediately terminates the process

- ❖ **Why kill is Important?**

Used when:

- Programs freeze
 - High CPU usage occurs
 - Unwanted processes must be stopped

- ❖ Syntax :

- kill PID

3. Use `top` or `htop` to monitor system resources and processes.

➤ The `top` command shows real-time system activity.

❖ **Command**

- `top`

❖ **Information Displayed**

- CPU usage
- Memory usage
- Running processes
- Load average

❖ **Why `top` is Important?**

Helps monitor:

- System performance
- Resource consumption
- Heavy processes

❖ **Using `Htop` :**

➤ `htop` is an advanced and user-friendly version of `top`.

❖ **Install `Htop` :**

- `sudo apt install htop`

❖ **Advantages of `htop`**

- Colorful interface
- Easy navigation
- Better process management

4. Configure one of your lab COMPUTERS to boot to the CLI using `systemd`, and reboot to confirm that you were successful.

❖ **What is `system` ?**

`system` manages:

- Services
- Startup process
- Boot targets

❖ **What is CLI Boot?**

➤ CLI (Command Line Interface) mode starts Linux without the graphical desktop environment.

Benefits:

- Uses fewer resources
- Faster boot time
- Useful for servers

❖ **Step-by-Step Configuration**

Step 1: Check Current Default Target
Step 2: Change Boot Mode to CLI
Step 3: Reboot the System
Step 4: Confirm CLI Boot

After reboot:

- System opens command-line login screen
- No graphical desktop loads

❖ Why This is Important?

- CLI mode is commonly used in:
- Linux servers
- Cloud systems

❖ Troubleshooting environments

- It improves:
- Performance
- Security
- Resource management